

MUHAMMAD WAQAS

CIVIL ENGINEER | BIM & VDC SPECIALIST | APPLIED AI & LLM ENGINEER

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EXECUTIVE PROFILE

Results-driven Civil Engineer, Manager Estate (Technical) at REDAMCO (Ministry of Railways), and Senior Infrastructure Specialist with 15+ years of field execution, project management, and technical appraisal experience across public sector mega-railways (Pakistan Railways), GCC/Oman projects, and multi-story housing developments. Certified ISO 19650 Information Manager & BIM Specialist proficient in Revit, Navisworks Manage (4D/5D), Civil 3D, Primavera P6, and Autodesk Construction Cloud. Applied AI & LLM Engineer specializing in AI Agentic workflows (N8N), generative architectural engines (AI-Architect), construction claim RAG intelligence platforms, and quantity surveying automation. Proven expertise in land monetization, project feasibility analysis, business development support, and digitally-enabled infrastructure delivery.

CORE COMPETENCIES & TECHNICAL MATRIX

Estate Management & Infrastructure

- Technical Project Appraisal & Execution Analysis
- Business Development & Asset Monetization Support
- Railway Land Development & Commercial Feasibilities
- Railway Infrastructure & Track Maintenance
- GIS Land Surveying & Encroachment Removal
- Quantity Surveying, IPCs & Contractual Claims

Applied AI & AEC Automation

- AI Architectural Layout Engine (AI-Architect)
- Construction Claim Analyzer (LLM + RAG)
- AI BOQ & Quantity Surveying Generator
- N8N AI Agentic Workflow Automation
- Python for AEC Systems & API Integration
- Machine Learning & Algorithmic Design

BIM, VDC & ISO 19650 Management

- Autodesk Revit (Architectural & Structural)
- Navisworks Manage (Clash & 4D/5D Simulation)
- Civil 3D (Corridor Design & Cut/Fill Earthworks)
- Primavera P6 (Scheduling, Baseline, EVM)
- ISO 19650 Information Management (Plannerly Certified)
- ACC / BIM 360, ReCap Pro, Twinmotion, ETABS

Software & Web Development

- Python, HTML5, CSS3, JavaScript
- Git / GitHub Version Control & Repositories
- Rest API Development & Web Dashboards
- Full-Stack Web App Development
- Microsoft Project & Advanced MS Office
- Technical Reporting & Project Governance

FEATURED AI & DIGITAL AEC INNOVATION PROJECTS

AI-Architect — Generative AI Floorplan & CAD/BIM Engine

Architectural & AI Software Project | GitHub Repo: github.com/engrwaqas034-AI/AI-Architect

- Developed an end-to-end Python & Web AI engine that automates 2D spatial layouts, boundary envelope math, setback compliance, and floor plan generation.
- Integrated real-time parametric SVG rendering, multi-floor space allocations, and direct export pipelines for CAD/BIM workflows.
- Designed custom CSP (Constraint Satisfaction Problem) solvers to eliminate room overlap and optimize natural lighting and ventilation.

Construction_Claim_Analyzer — LLM + RAG Claim Intelligence System

AI & Legal/Contractual Engineering Project | GitHub Repo: github.com/engrwaqas034-AI/Construction_Claim_Analyzer

- Engineered an LLM and Retrieval-Augmented Generation (RAG) system specialized in evaluating construction delay claims and Extension of Time (EOT) requests.
- Automated cross-referencing of project baseline schedules, daily site logs, and FIDIC contractual clauses for rapid dispute resolution.

- Provides automated claim vulnerability scoring and delay attribution matrices to assist claims consultants and project managers.

Government Primary School KPK — Complete 3D/4D/5D BIM Project

BIM & VDC Infrastructure Project | BIM PAK / BRAINS Institute Peshawar

- Led complete BIM lifecycle modeling in Revit (Architectural & Structural coordination) for an educational facility in KPK.
- Executed 4D construction sequencing in Navisworks Manage linked with Primavera P6 5D cost and 12-month schedule analysis.
- Produced high-fidelity Twinmotion photorealistic walkthroughs and BOQs, reducing on-site rework by 15%.

PROFESSIONAL EXPERIENCE (15+ YEARS)

MANAGER ESTATE (TECHNICAL) | REDAMCO — MINISTRY OF RAILWAYS

Rawalpindi / Islamabad | Dec 2025 – Present | Scope: Public Sector Corporate & Asset Management

- Analyze technical feasibilities, execution strategies, and structural/infrastructural appraisals for major Ministry of Railways estate projects.
- Assist the Senior Manager Estate and Business Development in strategic land monetization, commercial proposals, joint ventures, and revenue optimization.
- Perform technical risk assessments, lease compliance audits, and engineering evaluations on railway land assets across Pakistan.
- Implement digital project monitoring, GIS asset mapping, and BIM workflows to streamline real estate asset performance and commercial delivery.

INSPECTOR OF WORKS / INFRASTRUCTURE ENGINEER | PAKISTAN RAILWAYS

Rawalpindi, Pakistan | May 2018 – 09th Dec 2025 (91 Months) | Scope: Public Sector Client / Infrastructure

- Planned, executed, and monitored major railway infrastructure projects including track renewals, station rehabilitation, and bridge maintenance.
- Conducted detailed land and topographic surveys using GIS systems, removed ROW encroachments, and managed railway asset safety.
- Prepared cost estimates, engineering budgets, technical drawings, structural plans, and liaised with external departments for statutory approvals.
- Coordinated multi-disciplinary traffic and civil engineering operations to ensure zero-disruption train transit during site execution.

SITE ENGINEER (CIVIL) | SHAH ZAMAN CONSTRUCTIONS PVT LTD

Rawalpindi, Pakistan | Mar 2017 – Jan 2018 (10 Months) | Scope: DHA Emaar Housing Society Infrastructure Project

- Prepared and submitted Interim Payment Certificates (IPCs), variation orders, and extension of time (EOT) claims.
- Managed material procurement, Bar Bending Schedules (BBS), material demand lists, and BOQ compliance.
- Maintained detailed project execution schedules using Primavera P6, ensuring progress monitoring within budget constraints.
- Oversaw on-site quality control testing, verified subcontractor billings against completed works, and maintained contract documents.

SITE ENGINEER (CIVIL) | CONSTRUCTION INVADAR PVT LTD

Rawalpindi, Pakistan | Oct 2013 – Jul 2016 (33 Months) | Scope: Housing Directorate GHQ (Askari 5, 11 & 14 Projects)

- Supervised construction of 2xG+4 residential flats, luxury villas, internal road networks, and infrastructure works.
- Managed comprehensive site documentation including IPCs, Running Account Bills (RARs), deviation orders, and BBS.
- Enforced quality compliance through rigorous concrete testing, structural inspection, and batching plant coordination.
- Evaluated subcontractor performance and maintained progress schedules to achieve targeted milestones.

SITE ENGINEER (CIVIL) | AL RAAK INTERNATIONAL LLC

Muscat & Sohar, OMAN | May 2011 – Sep 2013 (28 Months) | Scope: Sohar Steel Mill Structure & Residential Villas (GCC Experience)

- Executed heavy frame structure building works for Sohar Steel Mill and multiple luxury residential villas in Oman.
- Oversee RCC works including slope protection, access ramps, concrete pavements, and precast element erection.
- Managed material procurement, layout marking, concrete batching plant delivery, and earthwork grading precision.

SUB ENGINEER (CIVIL) | AL SUBHAN CORPORATION

Rawalpindi, Pakistan | May 2010 – Jan 2011 (8 Months) | Scope: G+3 Residential Flats near CMH Rawalpindi

- Supervised structural concrete & steel works, site layout surveys, labor allocation, and material inventory.
- Prepared detailed work schedules, conducted joint inspections with consultants, and processed subcontractor invoices.

SITE SUPERVISOR (CIVIL) | IKSA CONSTRUCTIONS

Rawalpindi, Pakistan | Jan 2010 – May 2010 (4 Months) | Scope: Multistorey Ghakkar Plaza Commercial Project

- Supervised site layout marking, concrete pouring, steel reinforcement verification, and daily workforce output.

EDUCATION & ACADEMIC QUALIFICATIONS

Applied AI & LLM Engineering — Edersity Islamabad (2025 – 2026)

Specialization in AI Agentic Workflows, LLM RAG Architecture, and AI-driven Automation.

Post Graduate Diploma (PGD) in BIM & Construction Management — BRAINS Institute Peshawar (2025)

Specialization in Revit, Navisworks Manage (4D/5D), ISO 19650, Primavera P6, Civil 3D & VDC.

BS (Honors) in Civil Technology — Preston University Kohat, Islamabad Campus (2019)

3.6 GPA | RCC Design, Steel Structures, Highway & Tunnel Engineering, Theory of Structures.

Bachelor of Arts (Journalism & Education) — University of Sargodha (2015)

Graduated in Mass Communication, Media & Educational Methodologies.

Diploma of Associate Engineer (DAE Civil) — Government College of Technology Rasul (2009)

70% Marks | Quantity Surveying, Soil Mechanics, Construction Execution & RCC Design.

Secondary School Certificate (SSC Science) — APS & College Dhamial Camp Rawalpindi (2006)

72% Marks | Physics, Chemistry, Mathematics & Computer Science.

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

- ✓ Applied AI & LLM Engineer — Edersity Islamabad (2025/2026)
- ✓ Plannerly Certified Information Manager (ISO 19650 Level 1, 2, 3 Expert & Certification Ready)
- ✓ Mastering BIM with ISO 19650 — BIMPACC (2025)
- ✓ CanBIM Foundations Software Skills - Proficient (S4) — CanBIM / Skill Development Council Canada (2025)
- ✓ Autodesk Civil 3D Certification — Skill Development Council Canada (2025)
- ✓ Graduate Engineering Technologist — National Technology Council (NTC) Islamabad (2019)
- ✓ Diploma in Construction Management with Primavera P6 — Primavera Institute (2022)
- ✓ Revit Architecture Certification — PNY Trainings (2024)
- ✓ AutoCAD 2D & 3D Certification — E-Hunar (2024)
- ✓ Full-Stack Web Development Certification — Talha Tariq LMS (2025)
- ✓ AI Agentic Workflows & Automation — N8N Skool Community (2025)